

Problem Statement

Project Title

AI-Driven Plagiarism Intelligence System using IBM Granite and watsonx.ai

Official Problem Statement

Problem Statement No. 10 – AI-Driven Plagiarism Intelligence for Assignments

Academic institutions face increasing difficulty in detecting nuanced forms of plagiarism, especially when assignments are paraphrased or generated by AI tools. Traditional plagiarism detectors often rely on exact text matching and lack contextual understanding of writing style, instructor expectations, and historical submission patterns. As a result, sophisticated plagiarism frequently goes undetected.

The challenge is to develop an adaptive AI-powered system capable of analyzing assignment submissions, learning from historical data and instructor feedback, and identifying suspicious similarities with greater contextual awareness. The solution should enhance academic integrity by providing more accurate, explainable, and intelligent plagiarism detection.

Mandatory Technology: IBM Cloud Lite Services and IBM Granite Models.

Our Proposed Solution

The proposed **AI-Driven Plagiarism Intelligence System** leverages **IBM Granite** and **IBM watsonx.ai Runtime** to provide intelligent plagiarism detection beyond conventional similarity checking.

The system:

- Detects semantic, paraphrased, and AI-generated plagiarism.
 - Performs contextual similarity analysis using NLP techniques.
 - Generates AI-powered explanations with IBM Granite.
 - Produces plagiarism reports with confidence scores.
 - Supports PDF, DOCX, and TXT document analysis.
 - Tracks IBM Granite token usage for transparency.
 - Provides instructor-friendly feedback and recommendations.
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Objectives

- Detect semantic and AI-generated plagiarism.
- Improve contextual understanding of assignment similarity.

- Generate explainable plagiarism analysis.
 - Assist instructors with intelligent decision support.
 - Strengthen academic integrity using IBM AI technologies.
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Technology Stack

- IBM Granite Models
 - IBM watsonx.ai Runtime
 - IBM Cloud Lite
 - Python
 - Streamlit
 - Scikit-learn
 - TF-IDF & Cosine Similarity
 - PyPDF and python-docx
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Expected Outcome

The system provides an intelligent, explainable, and scalable plagiarism detection platform that helps educational institutions identify sophisticated plagiarism more accurately while supporting educators with AI-generated insights.